

Handwritten Karnaugh Map for a 4-variable function with variables b_1, b_0, b_3, b_2 .

	$b_1 b_0$ 00	01	11	10
$b_3 b_2$ 00	1	0	0	1
01	1	0	0	0
11	1	0	0	0
10	1	0	1	1

Groupings (circles) indicate prime implicants:

- Group 1: $b_1 b_0$ (top-left corner, covering cells (0,0), (0,1), (1,0), (1,1))
- Group 2: $b_3 b_2$ (top-right corner, covering cells (0,0), (1,0))
- Group 3: $b_1 b_0$ (bottom-left corner, covering cells (1,0), (1,1))
- Group 4: $b_3 b_2$ (bottom-right corner, covering cells (1,0), (1,1))

$$= \overline{b_1 b_0} + \overline{b_2 b_1} + \overline{b_3 b_2} b_1$$